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IMAGE AUTHENTICATION SYSTEM BASED ON TWO-DIMENSIONAL MATRIX CODES WITHOUT CONTAMINATION OF ORIGINAL IMAGES

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ABSTRACT

This paper proposes an authentication system without contamination of original images. The proposed system also realizes complete reconstruction of original signatures by introduction of 2-D matrix codes including error correction capability. Therefore, the system is suitable for the recent digital rights management.

1. INTRODUCTION

With the popular use of the Internet and the widespread of digital recording and storage, there have been a real need of digital rights management techniques. For this solution, many digital watermark schemes have been proposed [1]. However, since they directly embed signatures to original contents, the deterioration of their quality is inevitable. Further, the embedded signatures tend to be lost by changes of the original contents, such as compressions, blurring, etc. Therefore, we propose a new image authentication system without contamination of both original images and their signatures. In the proposed system, arbitrary signature data of the original image is expressed by a QR(Quick Response) code[1]. Further, a small number of parameters generating this code from the original image are saved to a database. Then, we can assign the signature to the original image without sacri cing its quality. In addition, the signature can be completely restored by error correction capability of the QR code even if the original image suffers from various changes. Therefore, the proposed system satis es the needs of the recent digital rights management.

2. PROPOSED AUTHENTICATION SYSTEM

Figure 1 gives the outline of the proposed system. The proposed system consists of two parts. One is a registration part of signatures. The other is a reconstruction part of signatures. The registration part gets an original image and its signature from users and converts the signature into a QR code. Since the QR code is capable of handling all types of data, arbitrary data format is available for the signature. Further, in the unit of generating authentication parameters, we calculate a small number of parameters, which reconstruct the QR code from the original image, and save them to a database. The details of this algorithm are explained later. Utilizing this new approach, we can assign the signature to the original image without sacri cing its quality.

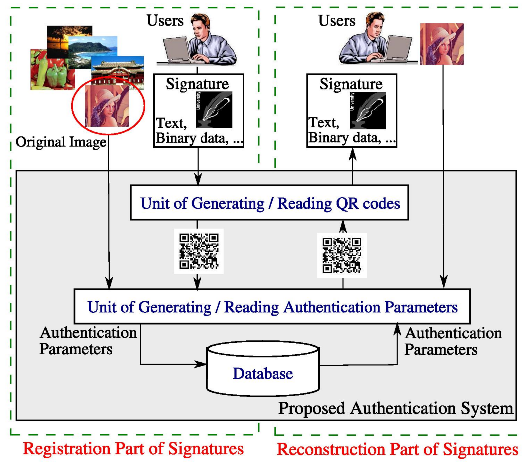

Fig. 1. Outline of the proposed authentication system.

In the reconstruction part of signatures, the proposed system gets an image desired to be authenticated and generates a QR code from the saved authentication parameters. Moreover, the obtained QR code is converted into the signature. In this way, we utilize the authentication parameters as a key to obtain the signature. Then, since any information cannot be known from only these parameters, the proposed system protects the user's anonymity. In addition, the QR code has an error correction capability. Therefore, even if it is damaged by some changes of the original image, such as compressions, blurring, etc, the signature can be completely reconstructed.

Further, we explain the details of the unit of the generating/reading authentication parameters. As shown in Fig. 2, this unit calculates the authentication parameters that reconstruct the QR code from the original image. First, since the QR code is a binary image, the original image is binarized by dithering [2] which is a common task for ink-jet printers.

[1]The QR code is developed by Denso-Wave in Japan. This code is a 2-D matrix code which is capable of handling all types of data, such as numeric and alphabetic characters, symbols, binary, control codes, etc.

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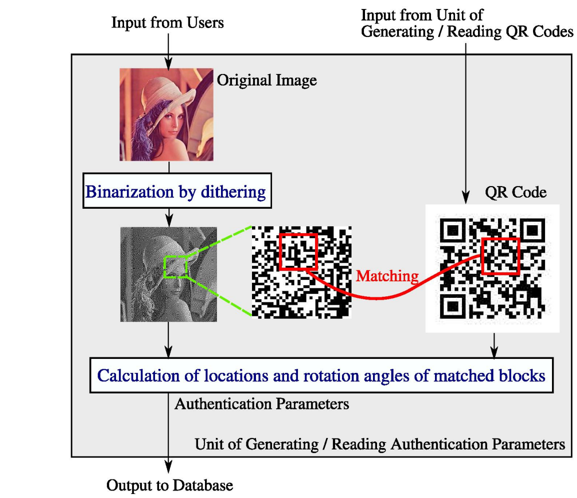

Fig. 2. Details of the unit of generating / reading authentication parameters.

Further, from the obtained binarized image, we select local areas that match the data area in the QR code. Then, the locations and the rotation angles of the selected local areas are saved to the database as the authentication parameters. Since the calculation scheme of these parameters is based on the fractal image coding [3], the amount of these data is very small. In this procedure, we also select the multiple local areas in order to realize the robust authentication over various changes of the original image. Note that when we authenticate the target image, the proposed system processes in the reverse direction and outputs the QR code.

Utilizing the above proposed system, we can realize the image authentication that ensures the quality of the original image and its signature. Therefore, the proposed system is suitable for the digital rights management.

3. SIMULATION RESULTS

This section shows the performance of the proposed system. We prepare 20 original images of various sizes. Further, for each image, the proposed system registers a signature of 50 alphabetic characters and authenticates 40 kinds of images. Figure 3 shows the rates of the correct pixels in the reconstructed QR codes. The error rates of the images (1-21) corresponding to the original are less than 4 %, so that we can reconstruct the original signature by the error correction capability of the QR codes. On the other hand, we cannot obtain the signature from the different images (22-40) because of the high error rates. Therefore, the proposed system realizes the accurate authentication.

Next, we verify the total amount of the authentication parameters. In the proposed system, the QR code is divide into 70 local blocks. Hence, for each block, if we calculate ve patterns of the parameters, x and y axis positions (5×2 bits) of matched blocks in the original image and their rotation angles (2 bits), the total amount is $(5{\times}2{+}2){\times}5{\times}70 = 4200$ bits (= 525 bytes). On the other hand, the amount of the JPEG encoded QR code is averagely about 5,785 bytes. Therefore, the proposed system is efficient from the viewpoint of the storage consumption. Further, for a image of size 2592×1944 pixels, the proposed system registers the signature at about 135 ms and reconstructs it at about 120 ms with a 3.2 GHz Pentium IV processor. Therefore, the proposed system is quite competent to apply to enormous amount of images.

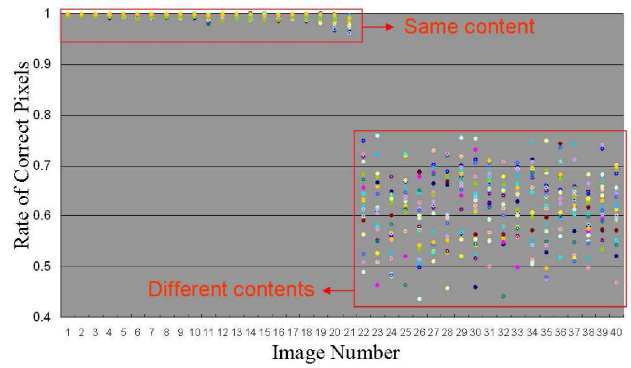

Fig. 3. The rates of the correct pixels in the reconstructed QR codes. The details of the image number are shown in Table 1.

Table 1. The relationship of the images and their numbers.

Numbers	Details of images
1	Original image
2-11	JPEG encoded images (quality 10,20,...,100)
12-16	Gaussian fi ltered images (radius 3,5,...,11)
17-21	Median fi ltered images (window size 3,5,...,11)
22-40	Different kinds of images

4. CONCLUSIONS

In this paper, we have proposed the image authentication system without contamination of original images. Further, the proposed system has also realized the complete reconstruction of the signatures by the introduction of the QR codes. The simulation results show the proposed system has the high authentication performance with low computation cost and storage consumption.

5. REFERENCES

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